Time Series — Solutions

Practice Exam — Week 2: ARMA Models and the PACF

EPFL, MATH-342 · Prof. S. C. Olhede

Problem 1 (5 points)

Statement.

(a) *State the definition of white noise and write down its ACVS γ_τ and ACF ρ_τ. Explain in one sentence why "uncorrelated" is weaker than "independent".* *(1.5 pts)*

(b) *Consider the MA(2) process*

$$X_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}.$$

Compute the mean and the autocovariance γ_τ for all *$\tau \in \mathbb{Z}$, and write down ρ_1 and ρ_2.* *(2.5 pts)*

(c) *For which lags τ does the ACVS of an MA(q) vanish? State the general rule and explain how it is used to read off the order q from an estimated ACF.* *(1 pt)*

Solution.

(a) $\{\varepsilon_t\}$ is *white noise* if it is a sequence of mean-zero, finite-variance, *uncorrelated* random variables: $\mathbb{E}[\varepsilon_t] = 0$, $\mathrm{Var}(\varepsilon_t) = \sigma^2 < \infty$, and $\mathrm{Cov}(\varepsilon_s, \varepsilon_t) = 0$ for $s \neq t$. Its second-order structure is

$$\gamma_\tau = \begin{cases} \sigma^2, & \tau = 0, \\ 0, & \tau \neq 0, \end{cases} \qquad \rho_\tau = \begin{cases} 1, & \tau = 0, \\ 0, & \tau \neq 0. \end{cases}$$

"Uncorrelated" only constrains the first two moments, whereas "independent" constrains the entire joint law; hence a white-noise process need not be i.i.d. (and need not be Gaussian).

(b) The constant mean is $\mathbb{E}[X_t] = \mathbb{E}[\varepsilon_t] - \theta_1 \mathbb{E}[\varepsilon_{t-1}] - \theta_2 \mathbb{E}[\varepsilon_{t-2}] = 0$. The filter taps are $(\psi_0, \psi_1, \psi_2) = (1, -\theta_1, -\theta_2)$. Using bilinearity and $\mathrm{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$, the ACVS is $\gamma_\tau = \sigma^2 \sum_j \psi_j \psi_{j+|\tau|}$ (only matching noise indices survive):

$$\begin{aligned}
\gamma_0 &= \sigma^2 \big(1 + \theta_1^2 + \theta_2^2\big), \\
\gamma_{\pm 1} &= \sigma^2 \big[(1)(-\theta_1) + (-\theta_1)(-\theta_2)\big] = \sigma^2\, \theta_1(\theta_2 - 1), \\
\gamma_{\pm 2} &= \sigma^2 \big[(1)(-\theta_2)\big] = -\sigma^2\, \theta_2, \\
\gamma_\tau &= 0 \qquad (|\tau| \geq 3).
\end{aligned}$$

Dividing by γ_0,

$$\rho_1 = \frac{\theta_1(\theta_2 - 1)}{1 + \theta_1^2 + \theta_2^2}, \qquad \rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2}.$$

(c) For an MA(q) with taps $\psi_0, \ldots, \psi_q$ one has $\gamma_\tau = \sigma^2 \sum_j \psi_j \psi_{j+|\tau|}$, and for $|\tau| > q$ no pair of taps overlaps (the filter has length $q+1$), so $\gamma_\tau = 0$ for $|\tau| > q$. Thus the **ACVS of an MA(q) cuts off after lag** q: the largest lag at which the (sample) ACF is still significant — beyond the $\pm 1.96/\sqrt{n}$ band — identifies the order q.

Problem 2 (5 points)

Statement.

Consider the AR(1) process $X_t = \phi X_{t-1} + \varepsilon_t$.

(a) *State the condition on ϕ for the process to be causal, and explain it in terms of the root of the AR polynomial $\Phi(z) = 1 - \phi z$.* *(1 pt)*

(b) *Assuming* $|\phi| < 1$*, derive the causal MA(*∞*) representation* $X_t = \sum_{k\geq 0}\psi_k\varepsilon_{t-k}$ *by iterating the recursion. Give the* ψ*-weights and verify* $\sum_k |\psi_k| < \infty$*.* *(2 pts)*

(c) *Using the representation in (b), show that*

$$\gamma_\tau = \frac{\sigma^2}{1-\phi^2}\,\phi^{|\tau|}, \qquad \rho_\tau = \phi^{|\tau|}.$$

Why does this ACF tail off *rather than cut off, and what does that say about identifying p from the ACF? (2 pts)*

Solution.

(a) The process is *causal* iff $|\phi| < 1$. The AR polynomial $\Phi(z) = 1 - \phi z$ has its single root at $z = 1/\phi$, and the causality condition "root outside the unit circle" reads $|1/\phi| > 1 \iff |\phi| < 1$.

(b) Iterating $X_t = \phi X_{t-1} + \varepsilon_t$, with $X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$, etc.:

$$\begin{aligned} X_t &= \varepsilon_t + \phi X_{t-1} = \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2 X_{t-2} \\ &= \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \cdots + \phi^{n-1}\varepsilon_{t-n+1} + \phi^n X_{t-n}. \end{aligned}$$

Since $|\phi| < 1$ and $\mathrm{Var}(X_{t-n}) = \gamma_0$ is constant, the remainder satisfies $\mathbb{E}[(\phi^n X_{t-n})^2] = \phi^{2n}\gamma_0 \to 0$, so it vanishes in mean square and

$$X_t = \sum_{k=0}^{\infty}\phi^k\varepsilon_{t-k}, \qquad \psi_k = \phi^k.$$

The weights are absolutely summable: $\sum_{k\geq 0}|\psi_k| = \sum_{k\geq 0}|\phi|^k = \dfrac{1}{1-|\phi|} < \infty$, confirming a valid causal MA(∞) representation.

(c) With $\mathbb{E}[X_t] = 0$ and $\mathrm{Cov}(\varepsilon_{t-k}, \varepsilon_{t-\tau-l}) = \sigma^2\mathbf{1}_{\{t-k=t-\tau-l\}}$, take $\tau \geq 0$:

$$\begin{aligned} \gamma_\tau &= \mathrm{Cov}\Big(\sum_{k\geq 0}\phi^k\varepsilon_{t-k},\ \sum_{l\geq 0}\phi^l\varepsilon_{t-\tau-l}\Big) = \sigma^2\sum_{l\geq 0}\phi^{l+\tau}\phi^l \\ &= \sigma^2\phi^\tau\sum_{l\geq 0}\phi^{2l} = \frac{\sigma^2\,\phi^\tau}{1-\phi^2}, \end{aligned}$$

where the surviving terms are those with $k = l + \tau$, and the geometric series converges since $\phi^2 < 1$. By symmetry $\gamma_\tau = \dfrac{\sigma^2}{1-\phi^2}\phi^{|\tau|}$, and $\rho_\tau = \gamma_\tau/\gamma_0 = \phi^{|\tau|}$.

The ACF $\phi^{|\tau|}$ decays geometrically but is never exactly 0 for $|\phi| \in (0, 1)$, so it *tails off* rather than cuts off. Consequently the order p of an AR cannot be read from the ACF; it is the **PACF** that cuts off at lag p (here $p = 1$).

Problem 3 (5 points)

Statement.

(a) *Consider the AR(2) process* $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ *with* $\phi_1 = \frac{1}{2},\ \phi_2 = -\frac{1}{4}$*. Using the causality conditions* $\phi_1 + \phi_2 < 1,\ \phi_2 - \phi_1 < 1,\ |\phi_2| < 1$*, verify that the process is causal.* *(1 pt)*

(b) *From the Yule–Walker equations* $\rho_\tau = \phi_1\rho_{\tau-1} + \phi_2\rho_{\tau-2}$ *(with* $\rho_0 = 1,\ \rho_{-1} = \rho_1$*), compute* ρ_1, ρ_2, ρ_3*. (2 pts)*

(c) *Using the Durbin–Levinson recursion*

$$\alpha_{1,1} = \rho_1, \qquad \alpha_{\tau,\tau} = \frac{\rho_\tau - \sum_{j=1}^{\tau-1}\alpha_{\tau-1,j}\,\rho_{\tau-j}}{1 - \sum_{j=1}^{\tau-1}\alpha_{\tau-1,j}\,\rho_j}, \qquad \alpha_{\tau,j} = \alpha_{\tau-1,j} - \alpha_{\tau,\tau}\,\alpha_{\tau-1,\tau-j},$$

compute $\alpha_1, \alpha_2, \alpha_3$ from the values in (b). Read off the implied model order and state the ACF/PACF identification rule it illustrates. *(2 pts)*

Solution.

(a) With $\phi_1 = \frac{1}{2},\ \phi_2 = -\frac{1}{4}$:

$$\phi_1 + \phi_2 = \tfrac{1}{4} < 1, \qquad \phi_2 - \phi_1 = -\tfrac{3}{4} < 1, \qquad |\phi_2| = \tfrac{1}{4} < 1.$$

All three causality conditions hold, so the AR(2) is causal (both roots of $\Phi(z) = 1 - \frac{1}{2}z + \frac{1}{4}z^2$ lie outside the unit circle).

(b) The Yule–Walker recursion $\rho_\tau = \phi_1\rho_{\tau-1} + \phi_2\rho_{\tau-2}$. At $\tau = 1$, using $\rho_{-1} = \rho_1$, we get $\rho_1 = \phi_1\rho_0 + \phi_2\rho_1 = \phi_1 + \phi_2\rho_1$, so

$$\rho_1 = \frac{\phi_1}{1-\phi_2} = \frac{1/2}{1+1/4} = \frac{1/2}{5/4} = \frac{2}{5}.$$

Then

$$\rho_2 = \phi_1\rho_1 + \phi_2 = \tfrac{1}{2}\cdot\tfrac{2}{5} - \tfrac{1}{4} = \tfrac{1}{5} - \tfrac{1}{4} = -\tfrac{1}{20},$$
$$\rho_3 = \phi_1\rho_2 + \phi_2\rho_1 = \tfrac{1}{2}\cdot\big(-\tfrac{1}{20}\big) - \tfrac{1}{4}\cdot\tfrac{2}{5} = -\tfrac{1}{40} - \tfrac{1}{10} = -\tfrac{1}{40} - \tfrac{4}{40} = -\tfrac{1}{8}.$$

(c) Durbin–Levinson with $\rho_1 = \frac{2}{5},\ \rho_2 = -\frac{1}{20},\ \rho_3 = -\frac{1}{8}$.

Lag 1: $\alpha_{1,1} = \rho_1 = \dfrac{2}{5}$.

Lag 2:

$$\alpha_{2,2} = \frac{\rho_2 - \alpha_{1,1}\rho_1}{1-\alpha_{1,1}\rho_1} = \frac{-\frac{1}{20} - \frac{2}{5}\cdot\frac{2}{5}}{1-\frac{2}{5}\cdot\frac{2}{5}} = \frac{-\frac{1}{20} - \frac{4}{25}}{1-\frac{4}{25}} = \frac{-\frac{21}{100}}{\frac{84}{100}} = -\frac{21}{84} = -\frac{1}{4}.$$

Update: $\alpha_{2,1} = \alpha_{1,1} - \alpha_{2,2}\,\alpha_{1,1} = \frac{2}{5} - \big(-\frac{1}{4}\big)\frac{2}{5} = \frac{2}{5} + \frac{1}{10} = \frac{1}{2}$.

Lag 3:

$$\alpha_{3,3} = \frac{\rho_3 - \alpha_{2,1}\rho_2 - \alpha_{2,2}\rho_1}{1-\alpha_{2,1}\rho_1 - \alpha_{2,2}\rho_2}.$$

The numerator is

$$-\tfrac{1}{8} - \tfrac{1}{2}\cdot\big(-\tfrac{1}{20}\big) - \big(-\tfrac{1}{4}\big)\cdot\tfrac{2}{5} = -\tfrac{1}{8} + \tfrac{1}{40} + \tfrac{1}{10} = -\tfrac{5}{40} + \tfrac{1}{40} + \tfrac{4}{40} = 0,$$

so $\alpha_{3,3} = 0$ (the denominator $1 - \frac{1}{2}\cdot\frac{2}{5} - (-\frac{1}{4})(-\frac{1}{20}) = 1 - \frac{1}{5} - \frac{1}{80} > 0$ is nonzero). Hence

$$\alpha_1 = \tfrac{2}{5}, \qquad \alpha_2 = -\tfrac{1}{4}, \qquad \alpha_3 = 0.$$

The PACF cuts off after lag 2 while the ACF in (b) tails off: this is the **AR(p) signature with $p = 2$**. Indeed the diagonal coefficients recover the AR coefficients, $\alpha_{2,1} = \frac{1}{2} = \phi_1$ and $\alpha_{2,2} = -\frac{1}{4} = \phi_2$. (Identification rule: PACF cuts after lag p, ACF tails off $\Rightarrow$ AR(p).)

Problem 4 (5 points)

Statement.

(Revision of Week 1: stationarity, the ACVS & ergodicity.) *Let $\{Y_t\}$ be i.i.d. with mean 0 and variance σ_Y^2, and define $X_t = Y_t + \frac{1}{2}Y_{t-1}$.*

(a) *Compute $\mathbb{E}[X_t]$ and the autocovariance γ_τ^X for* all *$\tau \in \mathbb{Z}$, give the lag-1 autocorrelation ρ_1, and explain in one sentence why $\{X_t\}$ is weakly stationary.* *(2 pts)*

(b) (Validity via positive semi-definiteness.) *A candidate sequence is a valid ACVS of some stationary process only if it is positive semi-definite. Decide whether $\gamma_0 = 1$, $\gamma_{\pm 1} = 0.8$, $\gamma_\tau = 0$ ($|\tau| \geq 2$) is a valid ACVS, by forming the 3×3 covariance matrix $\Gamma_3 = [\gamma_{j-k}]_{1\leq j,k\leq 3}$ and testing its definiteness.* *(2 pts)*

(c) (Mean ergodicity.) *Show that if* $\sum_\tau |\gamma_\tau| < \infty$ *then* $\operatorname{Var}(\bar{X}_n) \to 0$, *where* $\bar{X}_n = \frac{1}{n}\sum_{t=1}^n X_t$, *so* $\bar{X}_n$ *is a consistent (mean-ergodic) estimator of* $\mu = \mathbb{E}[X_t]$. *(1 pt)*

Solution.

(a) Since $\mathbb{E}[Y_t] = 0$, the mean is constant: $\mathbb{E}[X_t] = \mathbb{E}[Y_t] + \frac{1}{2}\mathbb{E}[Y_{t-1}] = 0$. The Y's are uncorrelated with variance σ_Y^2, so

$$\gamma_0^X = \operatorname{Var}(X_t) = \operatorname{Var}(Y_t) + \tfrac{1}{4}\operatorname{Var}(Y_{t-1}) = \Big(1 + \tfrac{1}{4}\Big)\sigma_Y^2 = \tfrac{5}{4}\sigma_Y^2,$$

$$\gamma_1^X = \operatorname{Cov}\big(Y_t + \tfrac{1}{2}Y_{t-1},\, Y_{t-1} + \tfrac{1}{2}Y_{t-2}\big) = \tfrac{1}{2}\operatorname{Var}(Y_{t-1}) = \tfrac{1}{2}\sigma_Y^2,$$

and $\gamma_\tau^X = 0$ for $|\tau| \geq 2$ (no shared Y index). Hence

$$\rho_1 = \frac{\gamma_1^X}{\gamma_0^X} = \frac{1/2}{5/4} = \frac{2}{5} = 0.4.$$

The mean is constant and γ_τ^X depends only on the lag τ (not on t), with $\gamma_0^X < \infty$; therefore $\{X_t\}$ is weakly stationary.

(b) The covariance matrix of (X_1, X_2, X_3) is the symmetric Toeplitz matrix

$$\Gamma_3 = \begin{pmatrix} 1 & 0.8 & 0 \\ 0.8 & 1 & 0.8 \\ 0 & 0.8 & 1 \end{pmatrix}.$$

Positive semi-definiteness requires *every* such covariance matrix Γ_n to have a non-negative determinant. Expanding along the first row,

$$\det\Gamma_3 = 1\cdot(1 - 0.8^2) - 0.8\cdot(0.8 - 0) + 0 = (1 - 0.64) - 0.64 = -0.28 < 0.$$

A positive-semi-definite matrix cannot have a negative determinant, so Γ_3 is *not* positive semi-definite (it has a negative eigenvalue). Hence the sequence is **not** a valid ACVS of any stationary process. (Intuitively a lag-1 correlation of 0.8 with zero correlation beyond lag 1 is "too strong": contrast part (a), whose genuine model only reaches $\rho_1 = 0.4$.)

(c) Expand the variance of the sample mean and group the $n - |\tau|$ pairs at each lag τ:

$$\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\sum_{s=1}^n\sum_{t=1}^n \operatorname{Cov}(X_s, X_t) = \frac{1}{n^2}\sum_{s,t}\gamma_{s-t} = \frac{1}{n}\sum_{\tau=-(n-1)}^{n-1}\Big(1 - \frac{|\tau|}{n}\Big)\gamma_\tau.$$

Bounding the weights by $0 \leq 1 - |\tau|/n \leq 1$,

$$\operatorname{Var}(\bar{X}_n) \leq \frac{1}{n}\sum_{\tau=-(n-1)}^{n-1}|\gamma_\tau| \leq \frac{1}{n}\sum_{\tau\in\mathbb{Z}}|\gamma_\tau| = \frac{C}{n} \xrightarrow[n\to\infty]{} 0,$$

with $C = \sum_\tau |\gamma_\tau| < \infty$ by hypothesis. Since $\mathbb{E}[\bar{X}_n] = \mu$, this means $\bar{X}_n \to \mu$ in mean square, i.e. $\bar{X}_n$ is a consistent (mean-ergodic) estimator of μ.

Problem 5 (5 points)

Statement.

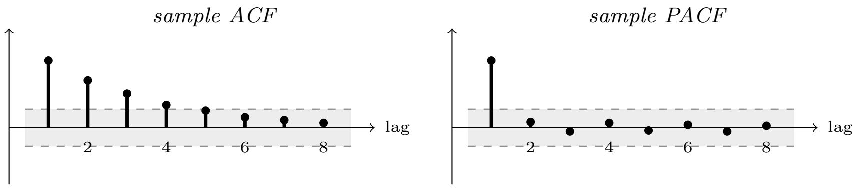
Model identification from a correlogram. *Three time series, each of length* $n = 100$, *produced the sample ACF and PACF shown (below, with each part). The dashed lines mark the* $\pm 1.96/\sqrt{n} \approx \pm 0.20$

*band and the lag-*0 *value (always* 1*) is omitted. For each series, name the most plausible model (white noise, MA(q), AR(p), ARMA(p, q)), give the order(s), and justify with the cut-off / tail-off rules.*

Solution.

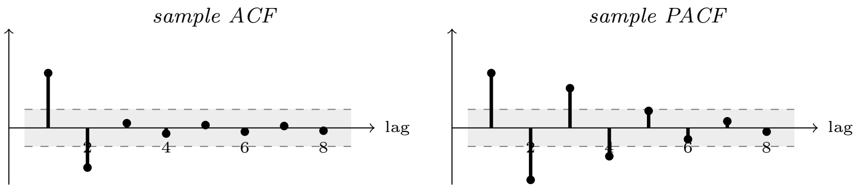
Identification rule (the punchline of the week). *ACF cuts off after lag q, PACF tails off* $\Rightarrow$ MA(q); *PACF cuts off after lag p, ACF tails off* $\Rightarrow$ AR(p); *both tail off* $\Rightarrow$ ARMA(p, q); *neither is significant* $\Rightarrow$ white noise. Here "cuts off" means the values drop inside the $\pm 1.96/\sqrt{n}$ band and stay there, while "tails off" means a gradual (geometric, possibly sign-alternating) decay.

(a)



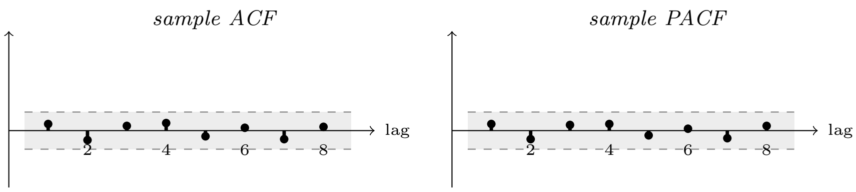
The PACF is significant only at lag 1 (≈ 0.71) and lies inside the band for every lag ≥ 2 — it *cuts off after lag* 1 — whereas the ACF decays geometrically $(0.71, 0.50, 0.36, 0.24, \cdots \approx 0.71^{\tau})$ and only *tails off*. PACF cut-off at lag 1 + ACF tail-off $\Rightarrow$ **AR(1)**, with $\phi \approx \rho_1 \approx 0.7 > 0$ (monotone positive decay).

(b)



The ACF is significant only at lags 1 and 2 (0.58, -0.42) and is inside the band for every lag ≥ 3 — it *cuts off after lag* 2 — whereas the PACF decays with alternating sign and only tails off. ACF cut-off at lag 2 + PACF tail-off $\Rightarrow$ **MA(2)**.

(c)



Every ACF and PACF value lies inside the ± 0.20 band, so nothing is statistically significant. There is no linear dependence to model: the series is consistent with **white noise**. Fitting any AR or MA structure here would be over-fitting pure noise. (Had *both* functions instead tailed off without either cutting off, the diagnosis would have been a mixed **ARMA(p, q)**.)